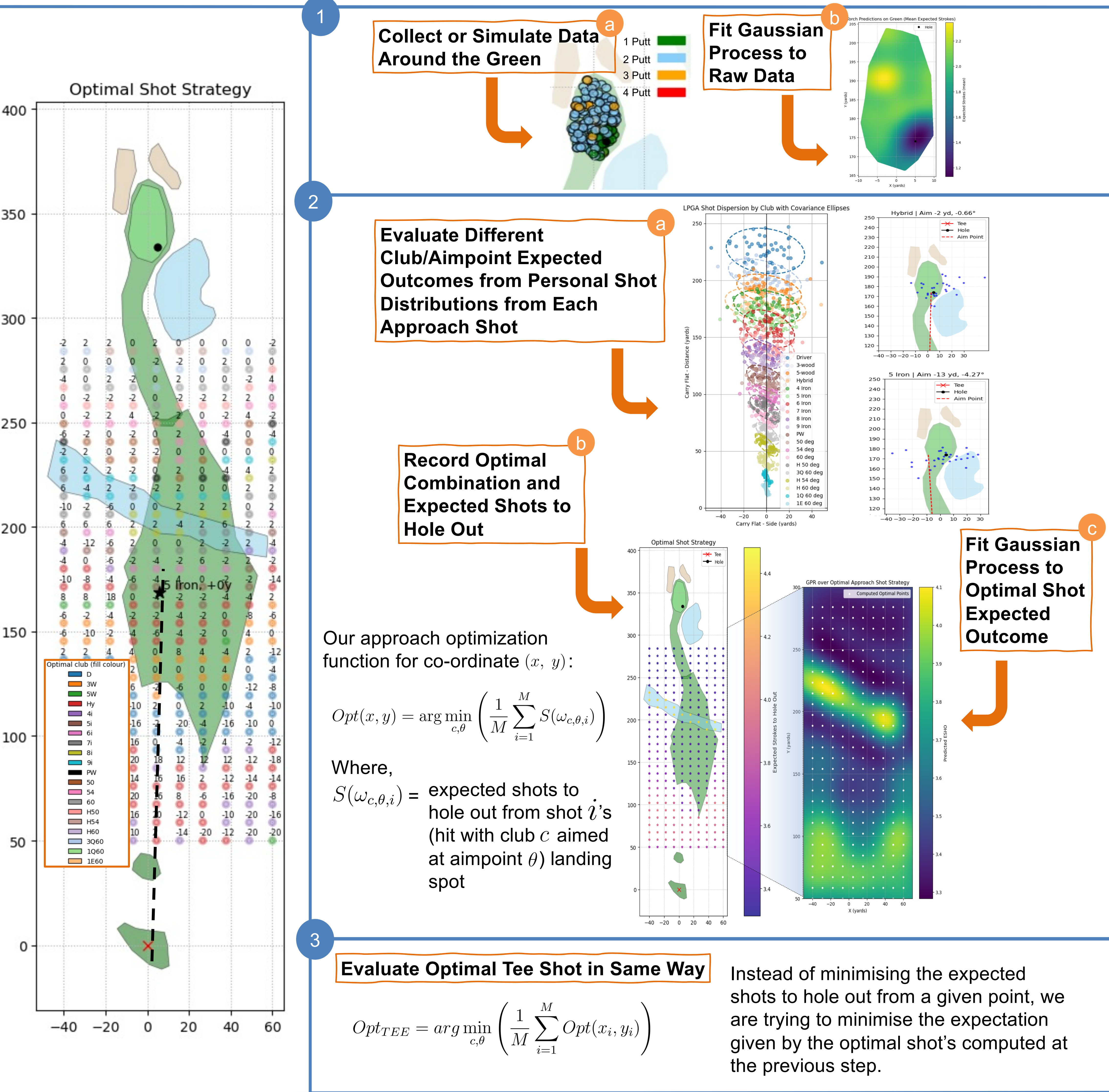


On Par: Leveraging Gaussian Process Regression for Perfect Golf

An Inside Out Approach for Simulating Optimal Par 4 Strategy



Introduction and Objectives

- Golf strategy is a sequential decision-making process.
- This project develops a simulation framework leveraging Gaussian Process Regression to model expected strokes to hole out and evaluate optimal club/aimpoint decision in a player-specific manner.
- Our project aimed to identify personalised strategies for optimising different objectives depending on the player's preferences or circumstances. We aimed to propose strategies:
 - Minimising expected strokes (risk averse)
 - Maximising birdie probability (risk prone)

Methodology

- Course geometry digitisation from GIS software.
- Realistic putting data generated by developing a mode implementing historical data and adjusting difficulty using multipliers.
- Player shot dispersion distributions derived from LPGA Trackman shot data, extended with simulated short game shots.
- Implementation of inside to out approach, building recursively on results, and evaluating results from Gaussian Process Regression.

Modeling Framework

Outlined above.

Results

Strategy Generation Under Different Loss Functions

Confirmed the feasibility of the proposed simulation framework in two different types of holes: par 3s and par 4s.

The different optimization mechanisms (birdie odds maximising vs expectation minimising) returned different outcomes. As expected, when minimising expected strokes, the model recommends more conservative strategies (less aggressive aimpoints such as laying up on tee shots). When maximising birdie probability, however, the model shifts toward aggressive choices (longer clubs and riskier lines). These differences are visible both in first tee shot selection and approach shot recommendations.

References

- Rasmussen & Williams (1995) "Gaussian Process for Regression"
- Lewis (2004) "Moneyball: The Art of Winning an Unfair Game"
- Broadie (2014) "Every Shot Counts"

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